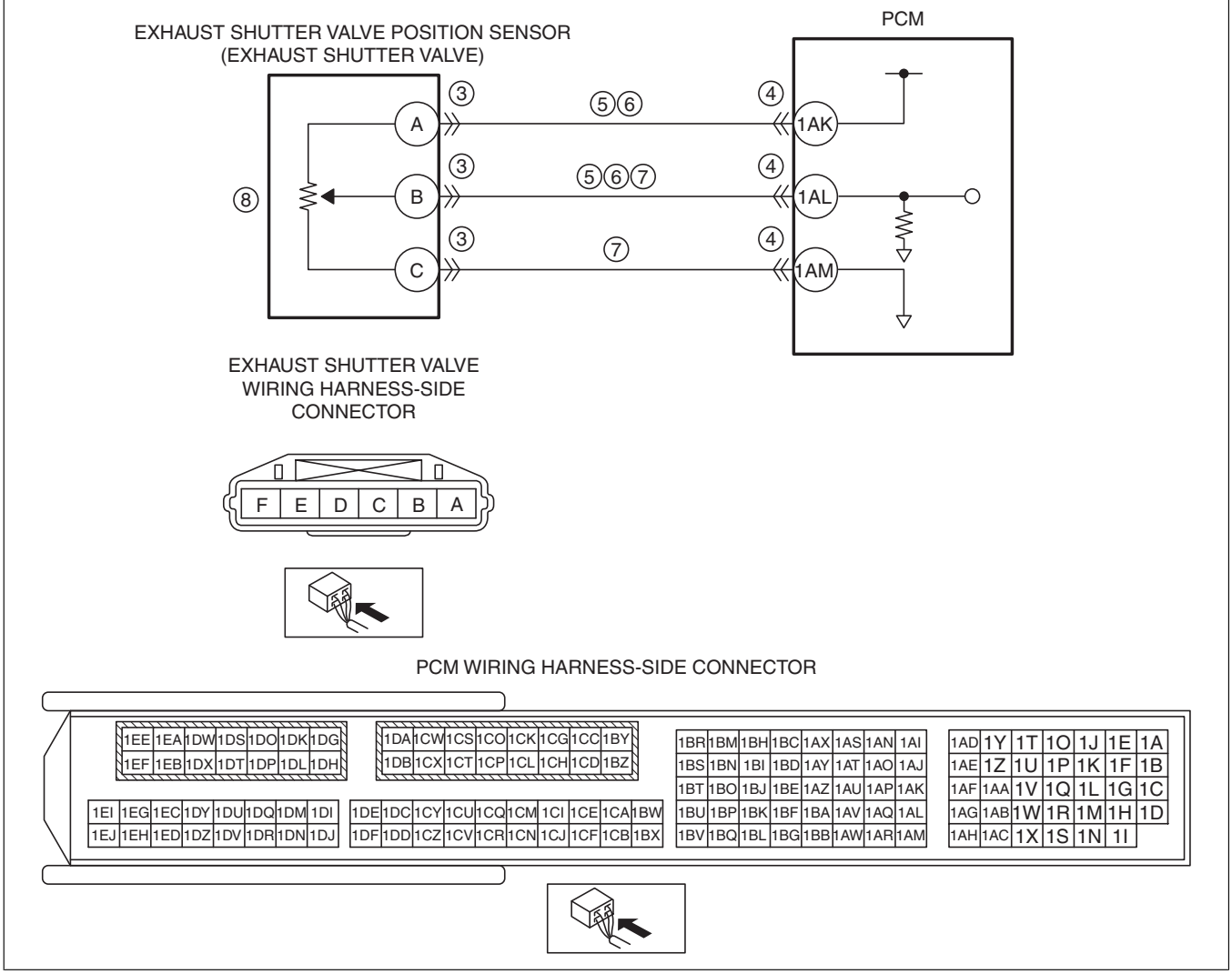
DTC P048D:00 [SKYACTIV-D 1.5]

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DTC P048D:00	Exhaust shutter valve position sensor circuit low input
DETECTION CONDITION	• If the input voltage at the PCM terminal 1AL is **less than 0.19 V** for **5 s**, the PCM determines that the exhaust shutter valve position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after ignition was switched on. — PCM internal processor malfunction DTC P0606:00 not detected. **Diagnostic support note** • This is a continuous monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque • Inhibits the EGR control
POSSIBLE CAUSE	• Exhaust shutter valve connector or terminals malfunction • PCM connector or terminals malfunction • Open circuit in wiring harness between the following terminals: — Exhaust shutter valve terminal A—PCM terminal 1AK — Exhaust shutter valve terminal B—PCM terminal 1AL • Short to ground in wiring harness between the following terminals: — Exhaust shutter valve terminal A—PCM terminal 1AK — Exhaust shutter valve terminal B—PCM terminal 1AL • Exhaust shutter valve position sensor signal circuit and ground circuit are shorted to each other • Exhaust shutter valve position sensor (built-into exhaust shutter valve) malfunction • PCM malfunction

DTC P048D: 00	Exhaust shutter valve position sensor circuit low input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related service information availability. • Is any related service information available?	Yes	Perform repair or diagnosis according to the available service information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT EXHAUST SHUTTER VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the exhaust shutter valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
5	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Exhaust shutter valve terminal A—PCM terminal 1AK — Exhaust shutter valve terminal B—PCM terminal 1AL • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust shutter valve terminal A—PCM terminal 1AK • Exhaust shutter valve terminal B—PCM terminal 1AL **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
6	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Exhaust shutter valve terminal A — Exhaust shutter valve terminal B • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust shutter valve terminal A—PCM terminal 1AK • Exhaust shutter valve terminal B—PCM terminal 1AL **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
7	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR SIGNAL CIRCUIT AND GROUND CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Inspect for continuity between exhaust shutter valve terminals B and C (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust shutter valve terminal B—PCM terminal 1AL • Exhaust shutter valve terminal C—PCM terminal 1AM **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 9.
		No	Go to the next step.
8	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR** • Reconnect all disconnected connectors. • Inspect the exhaust shutter valve position sensor. (See EXHAUST SHUTTER VALVE POSITION SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Inspect the exhaust shutter valve, then go to the next step. (See EXHAUST SHUTTER VALVE INSPECTION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.